Kia Karbasi

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AI/NLP researcher with a recent UCLA PhD and 10+ years across research, software development, and applied R&D. Recent work spans multi-agent LLM systems, text representations, sequential modeling, model interpretability, and empirical evaluation; earlier work includes large-scale network simulation, optimization, and resource allocation.

EXPERIENCE

University of California, Los Angeles - Pottie Lab, Los Angeles, CA

PhD Research, NLP and AI for Intelligent Tutoring Systems **2023–2026**

Advisor: Gregory J. Pottie

- Identified NLP/AI opportunities in intelligent tutoring systems by treating educational content and knowledge representations as language-rich modeling problems; developed new methods for content generation, knowledge representation, and student modeling.
- Conceived and developed multi-agent LLM systems for pedagogically aligned math-problem generation, improving difficulty matching by 12.5% and overall quality by 5.2% relative to each metric's best baseline.
- Reframed the interpretability of Deep Knowledge Tracing (DKT), a student-modeling method, showing its learned concept relationships align with prerequisite structure and help explain predictive behavior.
- Demonstrated DKT's learned concept relationships are not primarily semantic by constructing text-derived concept representations and finding only weak alignment with its learned relation structure.
- Proposed and evaluated embedding-based educational taxonomies by clustering question-text embeddings into alternative concept groupings; achieved best-observed relative DKT AUC gains of 6.4% and 3.8% over existing taxonomies on two benchmark datasets.
- Proposed and prototyped ReasonKT, an exploratory reasoning-based knowledge-tracing approach using Qwen with LoRA/GRPO and verifiable rewards from observed next-step outcomes, alongside o3 and GPT-4.1-mini reasoning experiments.

Wireless Communications Research **2016–2019**

Wireless Integrated Systems Laboratory | Co-PIs: Gregory J. Pottie and Babak Daneshrad

- Prototyped interference alignment, synchronization, and channel-estimation methods in MATLAB and on wireless hardware, demonstrating approximately 30–40 dB SINR gains per link; co-authored an IEEE MILCOM paper and presented the work to U.S. Navy research leadership.

The Aerospace Corporation, Los Angeles, CA

Senior Member of Technical Staff, Network Systems Department **2019–Present**

- Translated customer-defined mission questions into quantitative simulation studies across 30+ large-scale satellite-communications analyses, evaluating network and mission-level performance to support government decision-making under an active U.S. security clearance.
- Played a foundational role in developing the Space Networking Analysis Testbed (SNAT), establishing core simulation workflows, expanding the platform with new network-modeling capabilities, and later building major portions of its Key Performance Parameter (KPP) post-analysis suite.
- Developed optimization and machine-learning approaches for satellite communications, including genetic-algorithm ground-station placement and ML-based beam resource allocation, alongside broader software engineering and system testing.
- Led development of an NSF research proposal with two UCLA professors and one ASU professor, building the collaboration through internal PI outreach, organizing faculty technical talks, and cross-organizational research planning.

Visa Inc., Foster City, CA

Operations Systems Analyst **Dec 2014–Mar 2016**

Intern **Jun–Sep 2014**

- Designed and built an internal drag-and-drop platform for composing DevOps APIs into automated workflows, replacing manual processes and contributing to roughly 20% lower server downtime and 30% fewer operational work hours.

PUBLICATIONS

1. **Kia Karbasi**, Kaveh Karbasi, and Gregory J. Pottie. "Embedding-Based Educational Taxonomies." **MathNLP @ EMNLP 2026.**
2. **Kia Karbasi**, Kevin Hong, Mohammad Amin Samadi, and Gregory J. Pottie. "Multi-Agent Collaborative Framework for Math Problem Generation." **Educational Data Mining (EDM) 2025.**
3. Kevin Hong, **Kia Karbasi**, and Gregory J. Pottie. "Extracting Causal Relations in Deep Knowledge Tracing." **Educational Data Mining (EDM) 2025.**
4. Ehsan Keramat, Nicholas Kauffroath, **Kia Karbasi**, Heng Zhao, Babak Daneshrad, and Gregory J. Pottie. "Experimental Results for Low Overhead Frequency Offset Estimation in MANETs with Concurrent Transmission." **IEEE MILCOM 2017.**
5. **Kia Karbasi**, Mahdi Saeedi, Kevin Hong, and Gregory J. Pottie. "Are the KC Relations Learned by Deep Knowledge Tracing Semantic?" *Manuscript prepared for submission to the Journal of Educational Data Mining (**JEDM**).*

EDUCATION

University of California, Los Angeles

Electrical & Computer Engineering, PhD **2026**

Dissertation: On Applications of Natural Language Processing and Artificial Intelligence in Intelligent Tutoring Systems (Advisor: Gregory J. Pottie)

Electrical & Computer Engineering, MS **2018**

University of Maryland, College Park

Electrical Engineering, BS; GPA 3.9, cum laude. **2014**

SKILLS

AI/NLP & Modeling: Multi-agent systems; LLM agents; NLP; large language models (LLMs); generative AI; transformer models; prompt engineering; inference-time reasoning; model fine-tuning; PEFT/LoRA; reinforcement learning (GRPO); text embeddings; representation learning; deep learning; sequential modeling; model interpretability; clustering; dimensionality reduction; graph-based modeling.

Research & Evaluation: Experimental design; LLM evaluation; LLM-as-a-judge; evaluation harnesses; benchmarking; ablation studies; Bayesian hyperparameter optimization; statistical analysis; held-out evaluation; reproducible experimentation; experiment tracking; model-behavior and failure analysis.

Programming & ML Tools: Python; PyTorch; Hugging Face Transformers; scikit-learn; CUDA/GPU training; Weights & Biases; AutoGen; OpenAI API; Google Cloud; Git/GitHub; C++; SQL; MATLAB; Java.

Optimization & Simulation: Network modeling and simulation; optimization; resource allocation.

VENTURES, LEADERSHIP & ADDITIONAL EXPERIENCE

- **Founder, OneLanguage (2024):** Developed and submitted an Army SBIR proposal on low-resource multilingual machine translation across Malayo-Polynesian languages.
- **Research Lead, toPod (2025):** Developed and submitted an NSF SBIR/STTR proposal on using multimodal LLMs to transform scientific papers into accurate, accessible AI-generated podcasts.
- **Co-Founder/Developer, theBrodkast (2014):** Built a social-media link-sharing web/mobile application.
- **Teaching:** UCLA Teaching Assistant (selected quarters, **2016–2019 and 2024**).
- **Leadership & Service:** Founder, UMD Cinematheque Film Club (**2013–2014**; 1,500+ members); Board Member, UCLA Advanced Degree Consulting Club (**2019**); Graduate Student Representative, UCLA Academic Senate Graduate Council (**2017**); GSA Representative, Graduate Housing Oversight Committee (**2016**).
- **Investment Banking Analyst Intern, Venegas Capital (2017):** Financial modeling, valuation analysis, and industry research.
- **RF Field Technician (2010–2013):** Performed nationwide RF drive-testing and field measurement campaigns across 30 states for AT&T, Verizon, Sprint, and other major wireless carriers.
- **Interests:** Filmmaking; photography; literature and poetry; philosophy; guitar; travel.